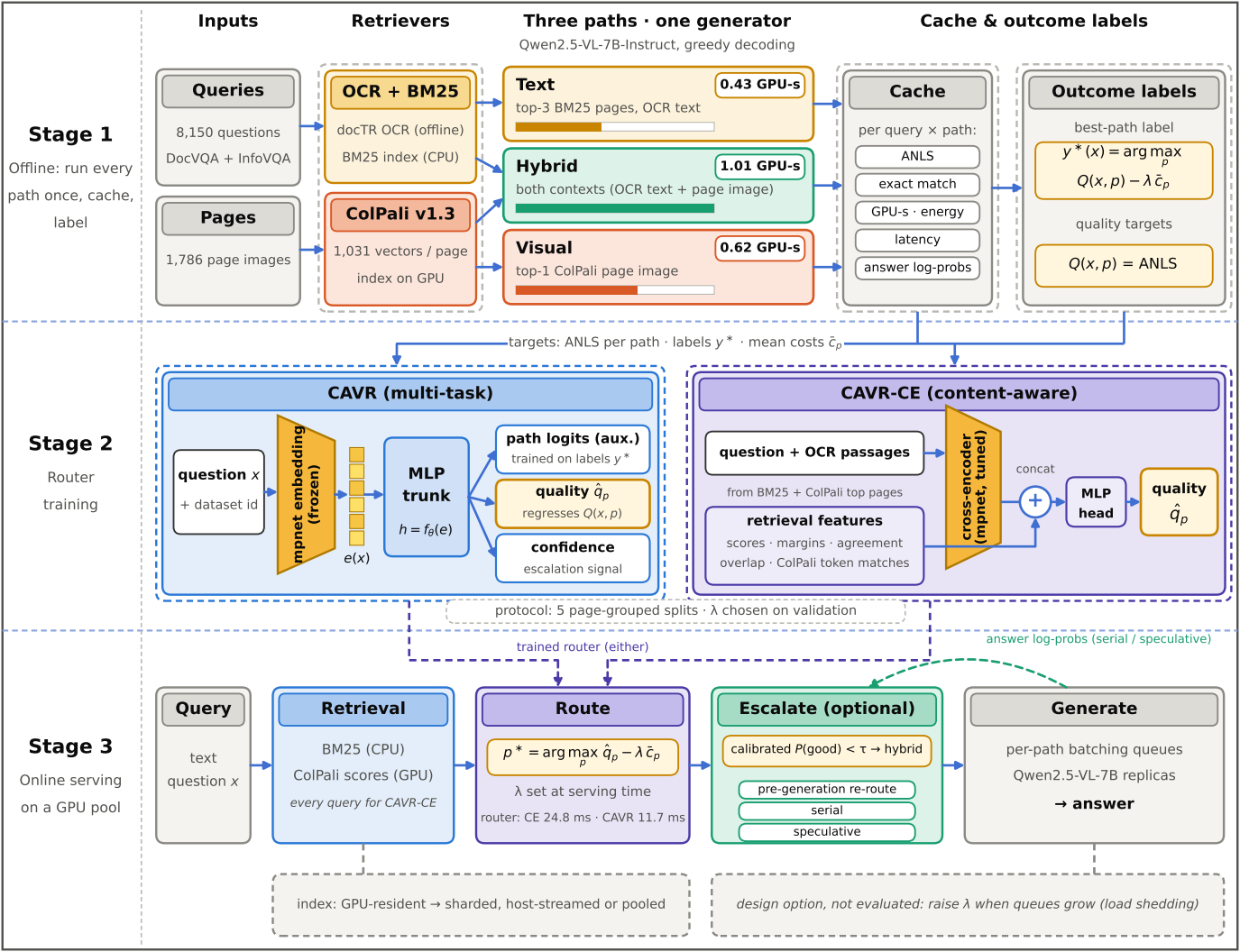

Inputs
Retrievers
Three paths · one generator
Qwen2.5-VL-7B-Instruct, greedy decoding
Cache & outcome labels
Stage 1
Offline: run every path once, cache, label
Queries
8,150 questions
DocVQA + InfoVQA
Pages
1,786 page images
OCR + BM25
docTR OCR (offline)
BM25 index (CPU)
ColPali v1.3
1,031 vectors / page
index on GPU
Text
0.43 GPU-s
top-3 BM25 pages, OCR text
Hybrid
1.01 GPU-s
both contexts (OCR text + page image)
Visual
0.62 GPU-s
top-1 ColPali page image
Cache
per query × path:
ANLS
exact match
GPU-s · energy
latency
answer log-probs
Outcome labels
best-path label
$y^*(x) = \arg\max_p Q(x,p) - \lambda \bar{c}_p$
quality targets
$Q(x,p)$ = ANLS
targets: ANLS per path · labels $y^*$ · mean costs $\bar{c}_p$
Stage 2
Router training
CAVR (multi-task)
question $x$
+ dataset id
mpnet embedding (frozen)
$e(x)$
MLP trunk
$h = f_\theta(e)$
path logits (aux.)
trained on labels $y^*$
quality $\hat{q}_p$
regresses $Q(x,p)$
confidence
escalation signal
CAVR-CE (content-aware)
question + OCR passages
from BM25 + ColPali top pages
retrieval features
scores · margins · agreement
overlap · ColPali token matches
cross-encoder (mpnet, tuned)
concat
MLP head
quality $\hat{q}_p$
protocol: 5 page-grouped splits · λ chosen on validation
trained router (either)
answer log-probs (serial / speculative)
Stage 3
Online serving on a GPU pool
Query
text question $x$
Retrieval
BM25 (CPU)
ColPali scores (GPU)
every query for CAVR-CE
Route
$p^* = \arg\max_p \hat{q}_p - \lambda \bar{c}_p$
λ set at serving time
router: CE 24.8 ms · CAVR 11.7 ms
Escalate (optional)
calibrated $P(\text{good}) < \tau \rightarrow$ hybrid
pre-generation re-route
serial
speculative
Generate
per-path batching queues
Qwen2.5-VL-7B replicas
→ answer
index: GPU-resident → sharded, host-streamed or pooled
design option, not evaluated: raise λ when queues grow (load shedding)